

LOK SABHA ELECTIONS 2019

ANALYSING THE ONLINE POLITICAL BATTLEFIELD IN INDIA

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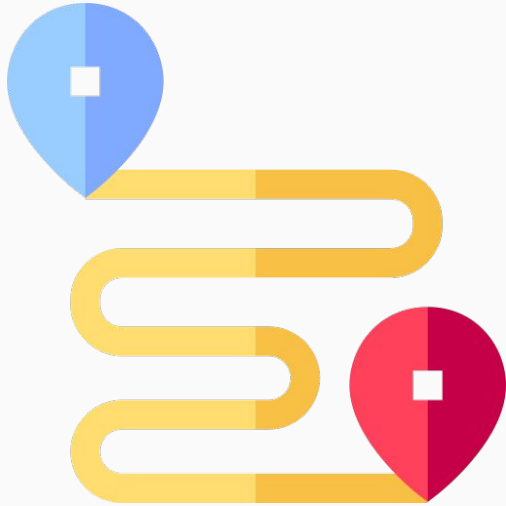
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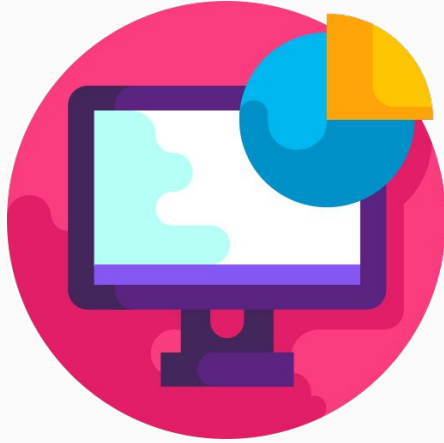


- Research Motivation
- Politics in India
- Data collection
- Preliminary Data Analysis
- Network analysis
- Discourse analysis
- Comparative analysis 2014 vs 2019
- Comparison with on-ground surveys
- Limitations and Future work



RESEARCH MOTIVATION

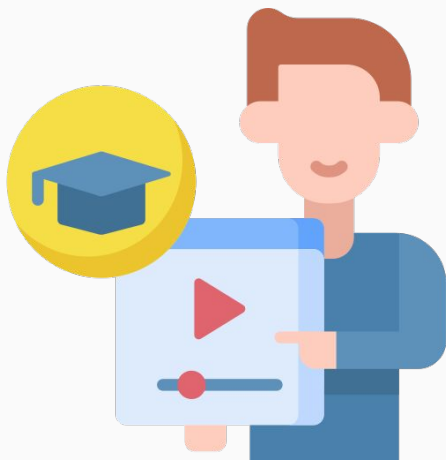
- Understanding Indian Political trends
 - 1947 - Present
 - How has politics shaped up?
 - How have frameworks operated?
 - Is there a pattern we see?
- Social Media & Elections
 - 2008 - Barack Obama
 - 2012 - Gujarat Elections
 - 2014 - Indian Lok Sabha Elections
- 2019 - A leap ahead
 - Manpower
 - Money
 - ECI Guidelines



2019 ELECTIONS

TECHNOLOGICAL POINT OF VIEW

- “Whatsapp elections”
- 30% population - access to social media
- 8 times more internet users than 2014
- 350 Million smartphone users
 - 2nd globally
 - Cheap data revolution (Jio)
 - 9.8 GB /month /smartphone
- 50,000 WhatsApp groups to spread campaign messages



CONTRIBUTIONS

- Studying the online conversations in 2019
 - How the conversations formed
 - How narratives were changed
- Understanding the social media presence of political parties and politicians
- Monitor Network of Verified politicians
 - How are they connected
 - How they interact
- Comparative analysis
 - 2019 vs 2014 - Online
 - 2019 - Online vs On-ground surveys



POLITICS IN INDIA



- 1947 - Independence
- 1952 - First Election (Single Party Dominance by INC)
- 1952 to 1963 - Nehru Era
- Weakening of INC's dominance
- Coming up of smaller parties
 - Region, Religion, Caste- Political Polarisation
- 1999 - INC went out of power for a full term
- Strong Two Party system
 - UPA - led by the INC
 - NDA - led by the BJP
- Ideal, shuffle of power

2 MAJOR POLITICAL PARTIES



Indian National Congress (INC)

- Agenda - Independence
- First Major Political Party
- Secular and center-Left aligned
- Leader of United Progressive Alliance
(UPA)

Bharatiya Janata Party (BJP)

- Representation of the Hindu Majority
- Political Wing of RSS
- Hindu Nationalist Volunteer Organisation
- First Major opposition to INC
- Largest political party by membership
- Leader of National Democratic Alliance(NDA)



- BJP marked a massive victory
- Major narratives that led to this -
 - Low popularity of left parties
 - Nationalism, national security, economic reforms
 - Lack of a “viable political alternative”



DATASET

45,177,116 TWEETS COLLECTED

From December 2018
to May 2019

2,150,179 ACTIVE USERS

2,585 HANDLES

Being monitored; Annotated with their political affiliation

1,012 POLITICIAN HANDLES

Contesting elections; extracted from ECI Website;
Out of 8,055 candidates, 1,012 on Twitter; 32%
Verified

1,252 VERIFIED POLITICIANS

Identified for Social Network Graph analysis



METHODOLOGY

- Understanding political discussions on Twitter
- Month-wise analysis (Jan-May 2019)
 - Party-wise Segregation
 - Focussing on INC and BJP
- LDA Topic Modelling
- Term-Frequency based Tag clouds
- Network Analysis - Gephi
- TF-IDF



PRELIMINARY ANALYSIS



MANIFESTO ANALYSIS

INC MANIFESTO



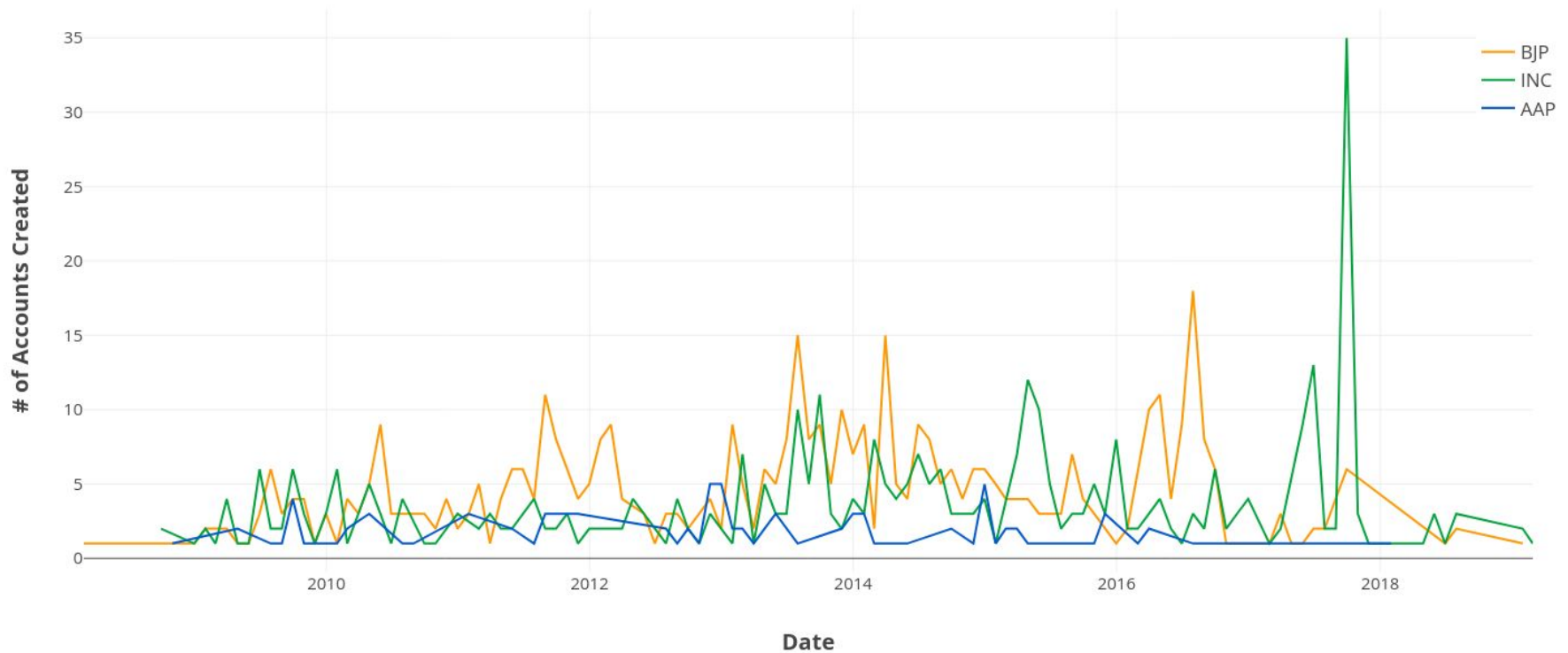
- Generic words - *government, development, growth, rural, bill, reforms*
- *Women - 33% Reservation*
- *Education - Double to 6% of GDP*
- *Right - Healthcare, Housing, Homestead*
- *Caste - Low mention*
- *National Security - Very low mention*

BJP - SANKALP PATRA



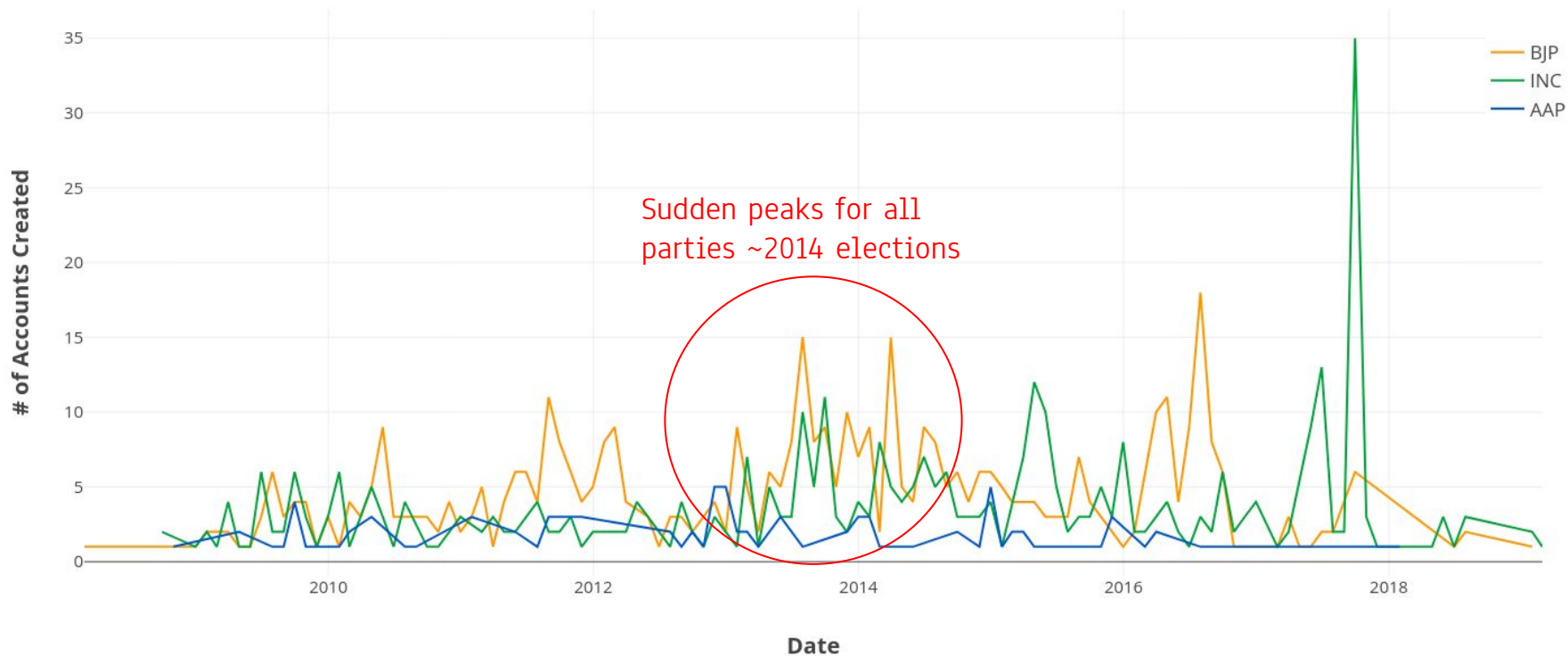
- Generic words - Development, Government, Country, five, efforts
- National Security - Not very high
- Women - Moderate mention (triple talaq)
- Low mention of issues like - farmers, jobs, health, infrastructure, education, employment, dalits etc.
- Highly criticised for unemployment, farmers distress

CREATION TIMES OF POLITICAL HANDLES



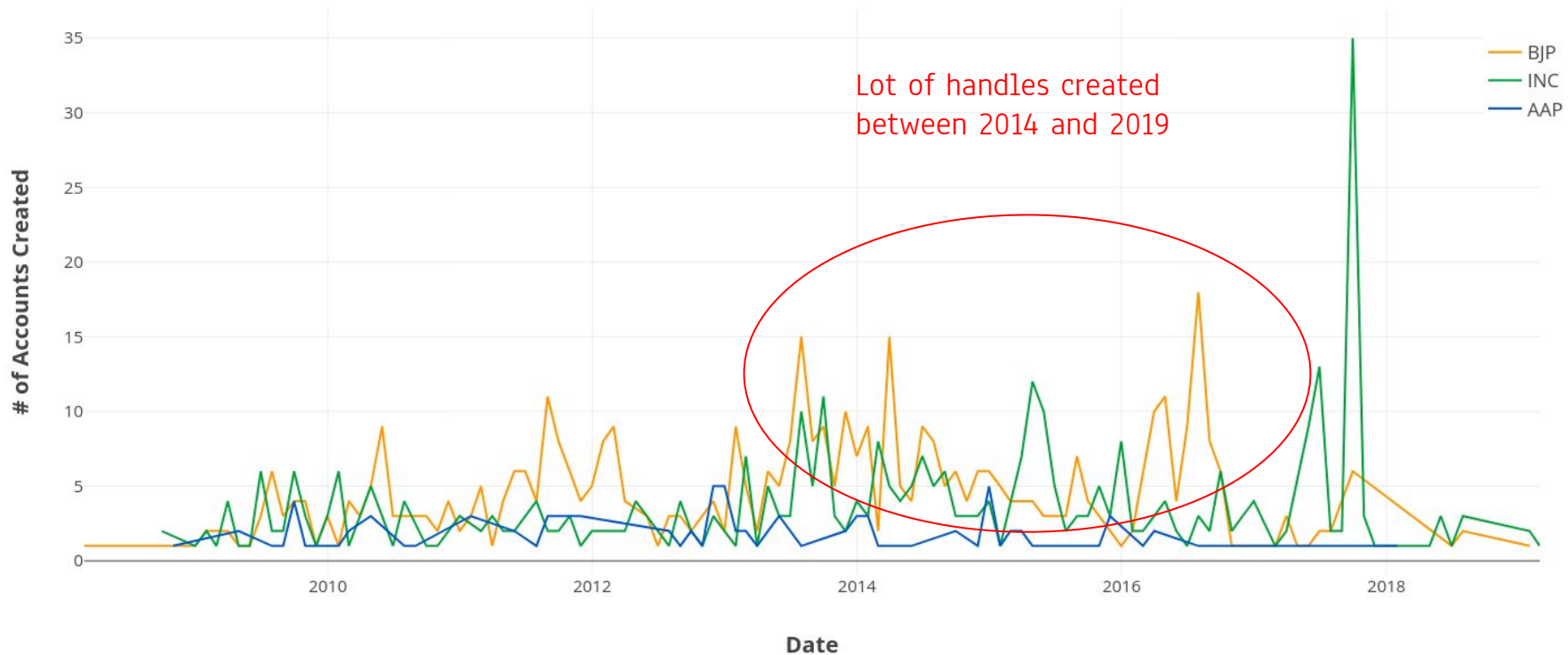
(as of March 2019)

CREATION TIMES OF POLITICAL HANDLES



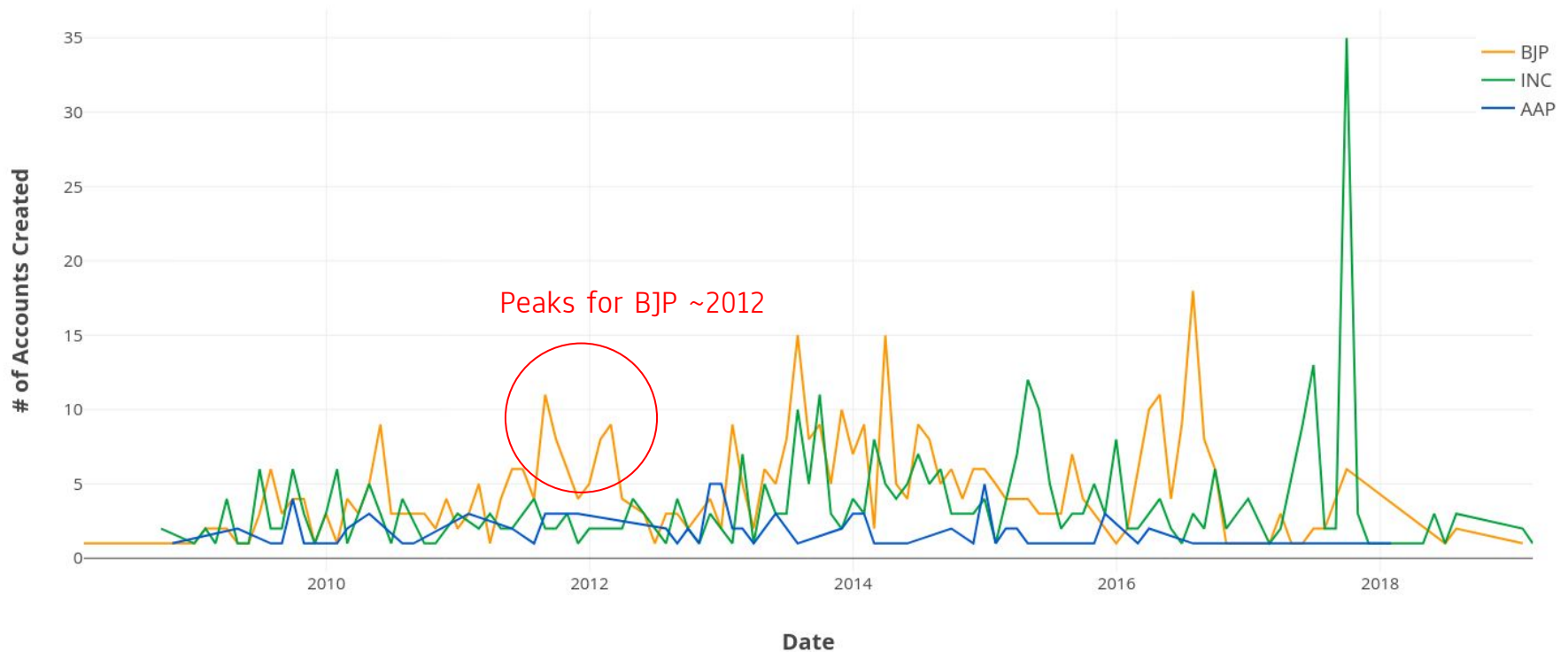
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CREATION TIMES OF POLITICAL HANDLES

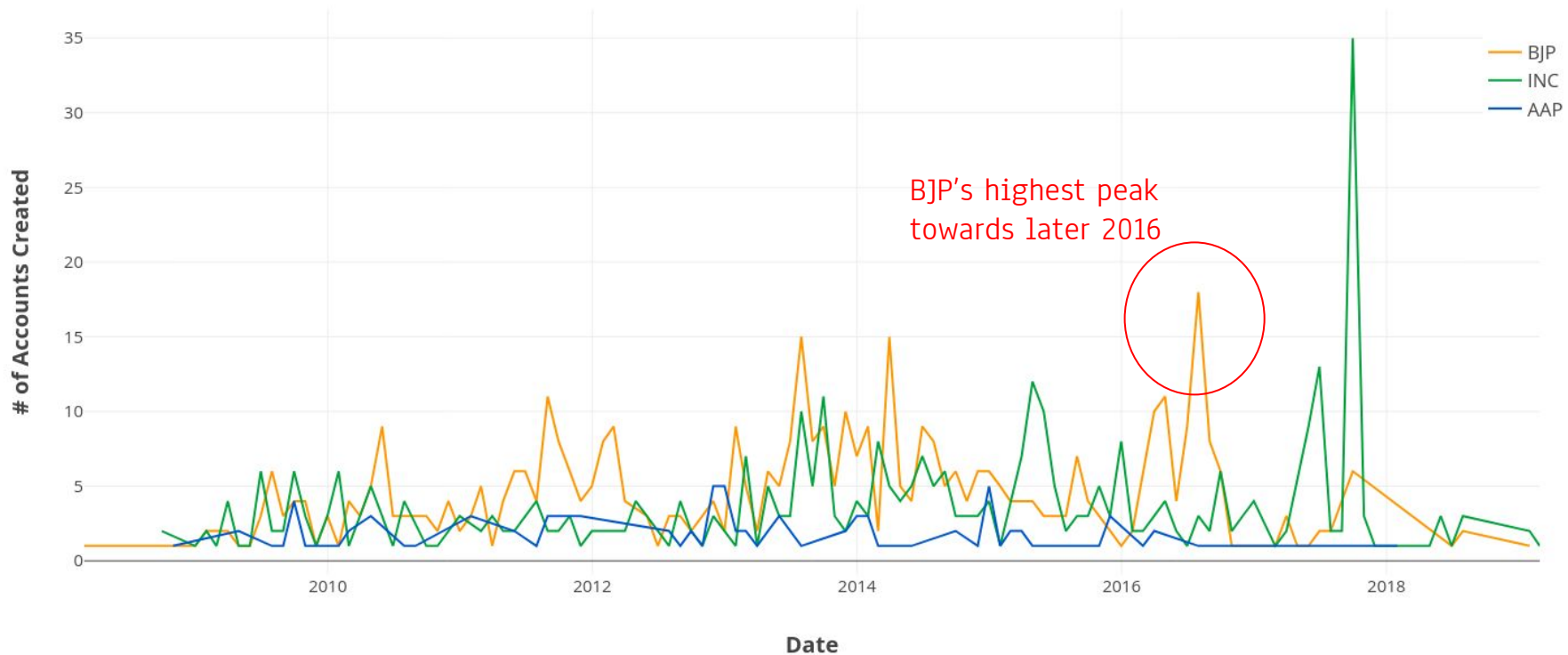


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CREATION TIMES OF POLITICAL HANDLES

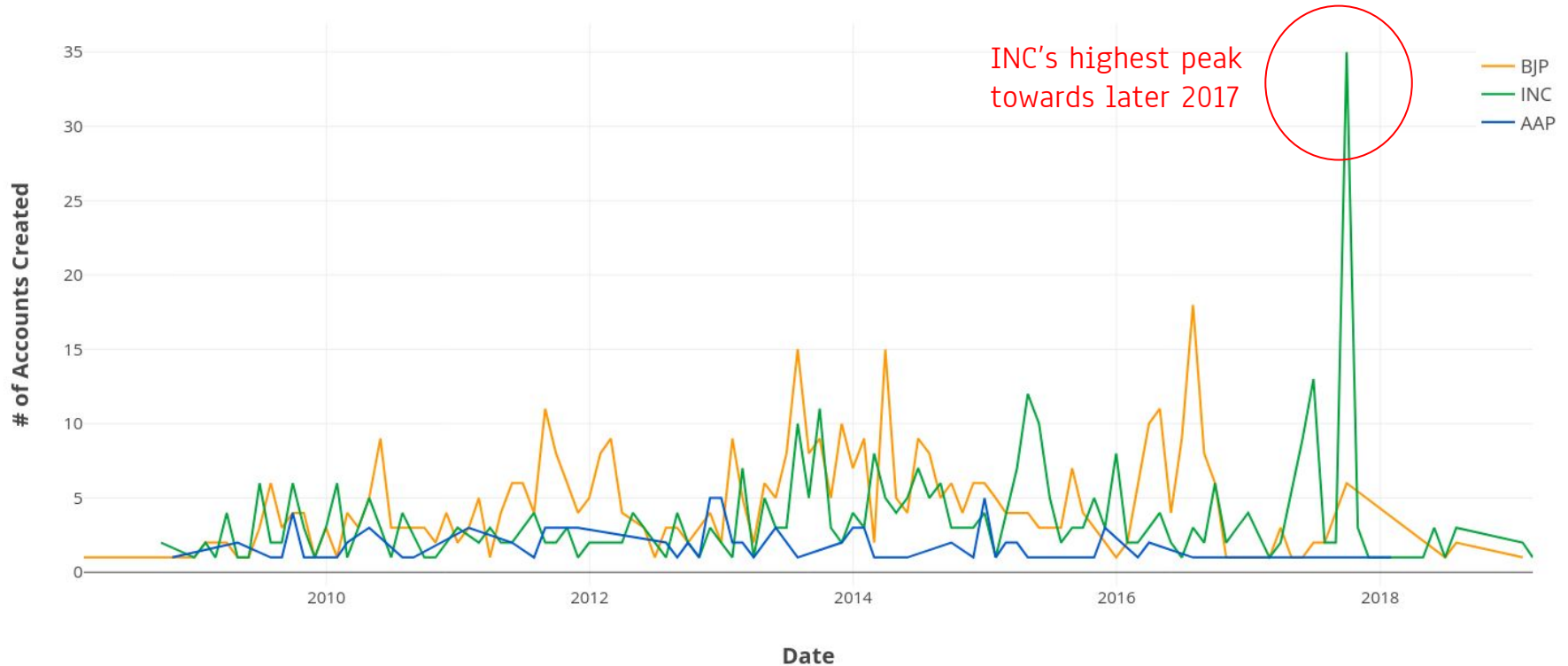


CREATION TIMES OF POLITICAL HANDLES



(as of March 2019)

CREATION TIMES OF POLITICAL HANDLES



(as of March 2019)

PARTY vs NUMBER OF HANDLES



(as of March 2019)

GENDER DISTRIBUTION



(as of March 2019)

NAMO vs RAGA

<i>ANALYSIS</i>	<i>NAMO</i>	<i>RAGA</i>
<i>ACCOUNT CREATED</i>	2009	2015
<i>FOLLOWERS</i>	50 MILLION	10.8 MILLION
<i>TWEETS</i>	5,219	684
<i>TIMES RETWEETED</i>	14,785,251	5,069,860
<i>TIMES FAVORITED</i>	61,312,161	16,396,994

(as of March 2019)

NAMO VS RAGA



A word cloud for Narendra Modi. The most prominent words are 'Government', 'BJP', 'nation', 'Thank', 'work', 'people', 'India', 'today', and 'state'. Other visible words include 'wonderful', 'Best wishes', 'congratulate', 'Odisha', 'May', 'tomorrow', 'proud', 'farmer', 'power', 'MannKiBaat', 'health', 'Citizen', 'poor', 'project', 'women', 'progress', 'well', 'Glad', 'watch', 'young', 'visit', 'life', 'Delhi', 'Karnataka', 'Chhattisgarh', 'leader', 'first', 'Happy', 'historic', 'effort', 'youth', 'development', 'working', 'Delighted', 'culture', 'world', 'several', 'country', 'PM narendramodi', 'student', 'rally', 'youngster', 'sector', 'part', 'special', 'Gandhi150', 'congratulations', 'Glad', 'young', 'visit', 'life', 'Delhi', 'Karnataka', 'Chhattisgarh', 'leader', 'first', 'Happy', 'historic', 'effort', 'youth', 'development', 'working', 'Delighted', 'culture', 'world', 'several', 'country', 'PM narendramodi', 'student', 'rally', 'youngster', 'sector', 'part', 'special', 'Gandhi150'.

NARENDRA MODI



A word cloud for Rahul Gandhi. The most prominent words are 'BJP', 'Karnataka', 'party', 'PM', 'people', 'today', 'Modi', and 'thank'. Other visible words include 'world', 'institution', 'great', 'stand', 'price', 'Happy', 'job', 'CBI', 'hatred', 'farmer', 'power', 'corruption', 'love', 'lie', 'meeting', 'question', 'Facebook', 'HAL', 'Indians', 'Govt', 'friend', 'act', 'President', 'Parliament', 'nation', 'salute', 'member', 'Delhi', 'justice', 'support', 'condolences', 'family', 'truth', 'RAFALE', 'Kerala', 'state', 'team', 'Anil Ambani', 'RafaleScam'.

RAHUL GANDHI

NAMO vs RAGA



NARENDRA MODI

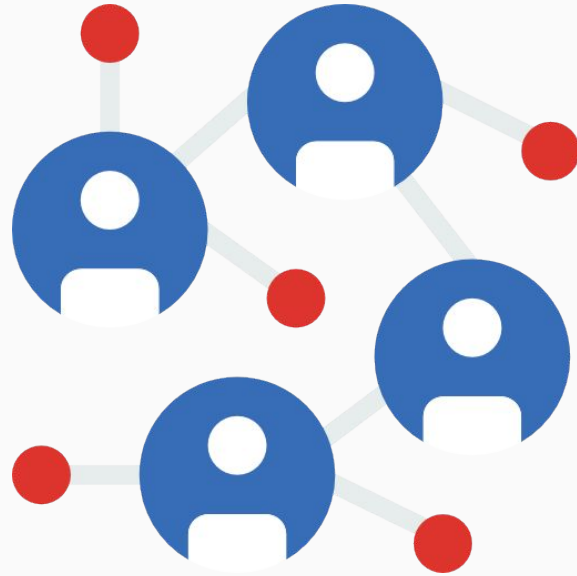
- No mention of *Women, youth, farmers*
- Engaging with people
- *Development, India, Government, Effort, work*
 - Praise government
- 'Good Vision'
- No negative words
 - Less criticism

NAMO vs RAGA

- Frequent mention of *PM, Modi, BJP*
- Issues - *Rafale scam, job, corruption, CBI, hatred, farmer, justice*
- Criticising the government
 - Leader of opposition



RAHUL GANDHI



NETWORK ANALYSIS

METHODOLOGY

- 2,585 Politically active handles
- 1,252 Verified handles; Authentic.
- Manual annotated
- Network analysis tool - Gephi
- Network visualisation + Network Metrics
 - In/Out degree, centrality, closeness, betweenness centrality, clustering coefficient, etc.

METHODOLOGY

How is the network built?

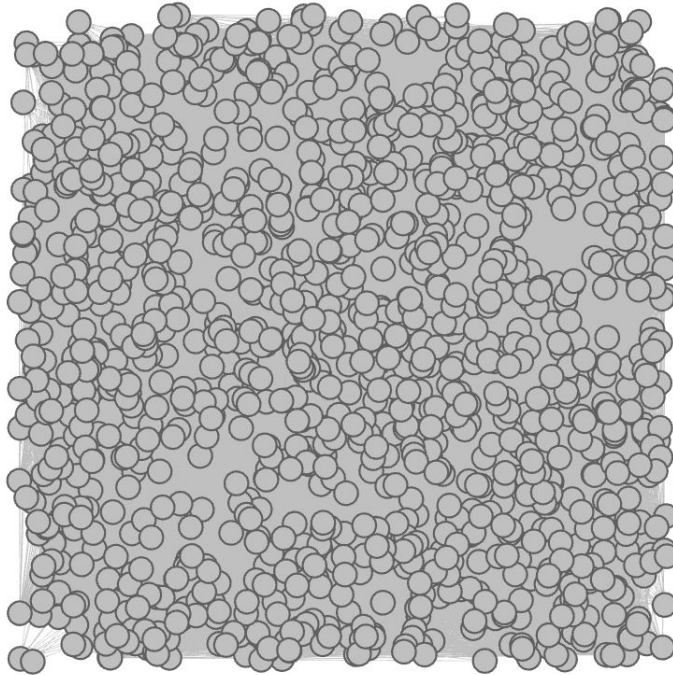
- Node - Verified political handle
- Node size - # of connections
- Node Color - Assigned Manually based on the party
- Edge - Follower/Following connected between 2 nodes
 - Clockwise curve - Outgoing (Following)
 - Anti-clockwise curve - Incoming (Follower)

METHODOLOGY

- Step - 1

Nodes of 1,252 Verified

Handles



METHODOLOGY

- Step - 2

Forming clusters or
communities in the
network



METHODOLOGY

- Step - 3

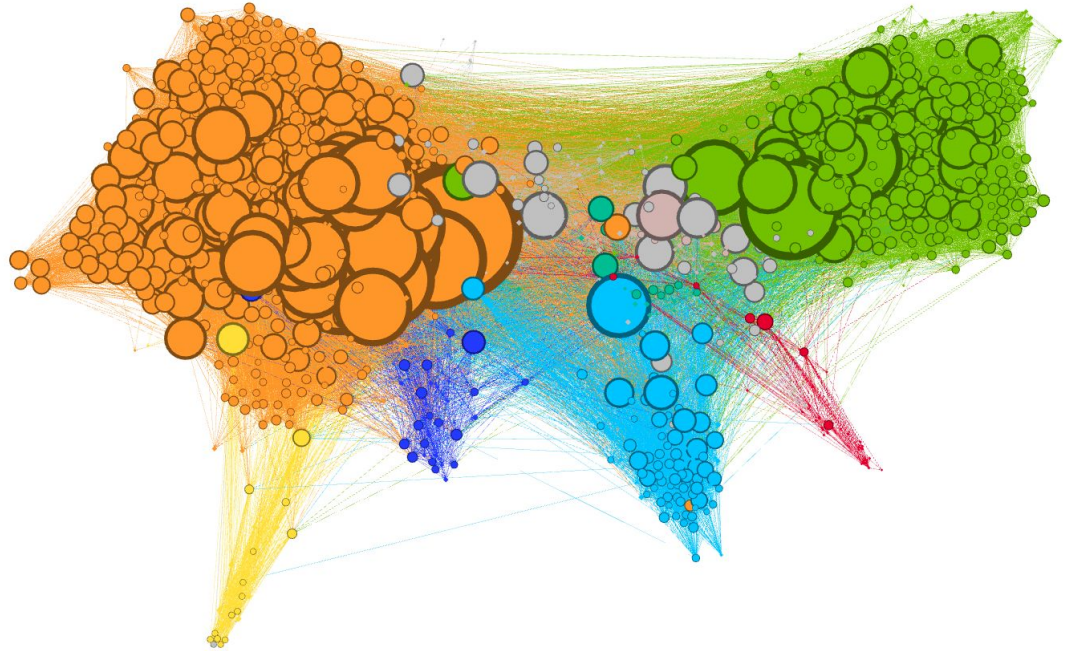
Assigning weights to
nodes based on Degrees



METHODOLOGY

- Step - 4

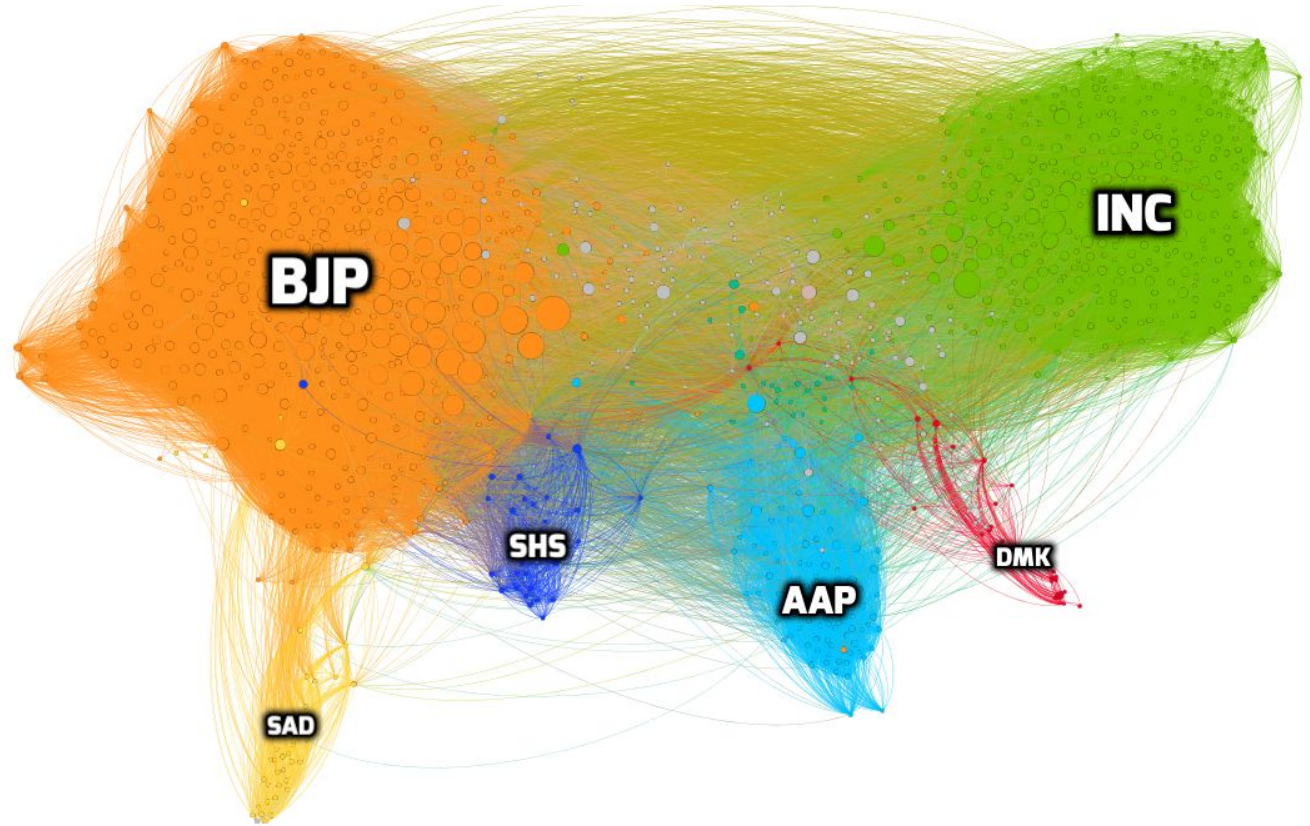
Assigning colors based on
Manual party annotations



UNDERSTANDING THE NETWORK VISUALISATION

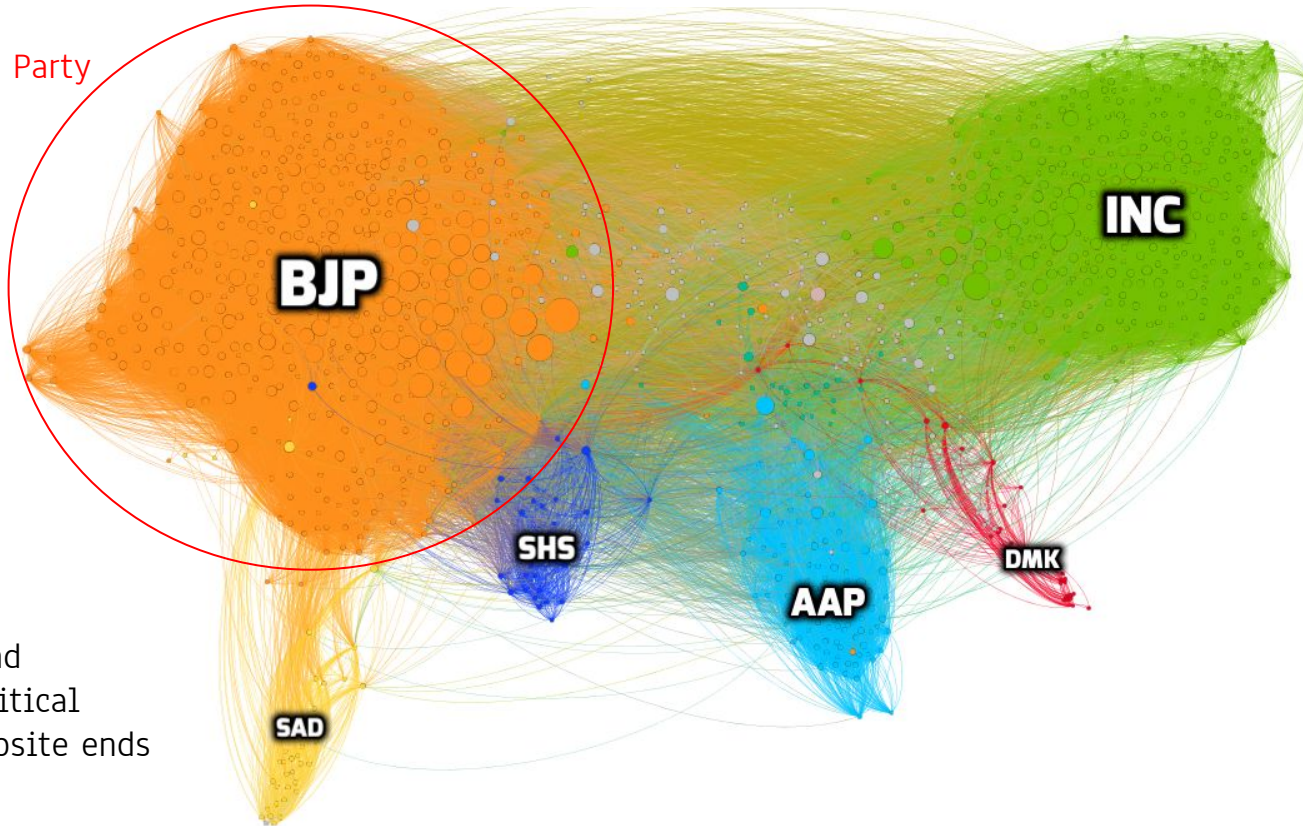
Clusters + Colors

=> Members of same
party in the same cluster



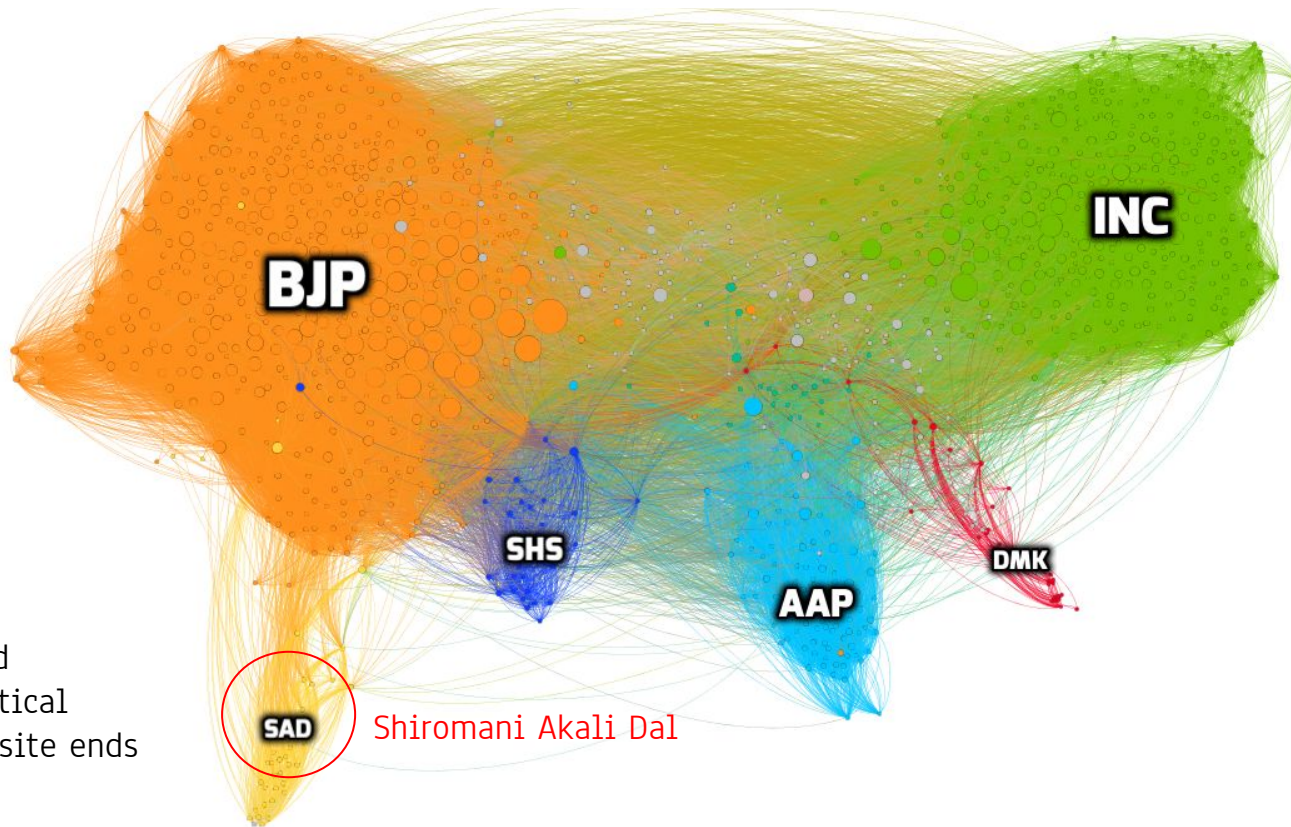
ANALYSING THE NETWORK VISUALISATION

Bharatiya Janata Party

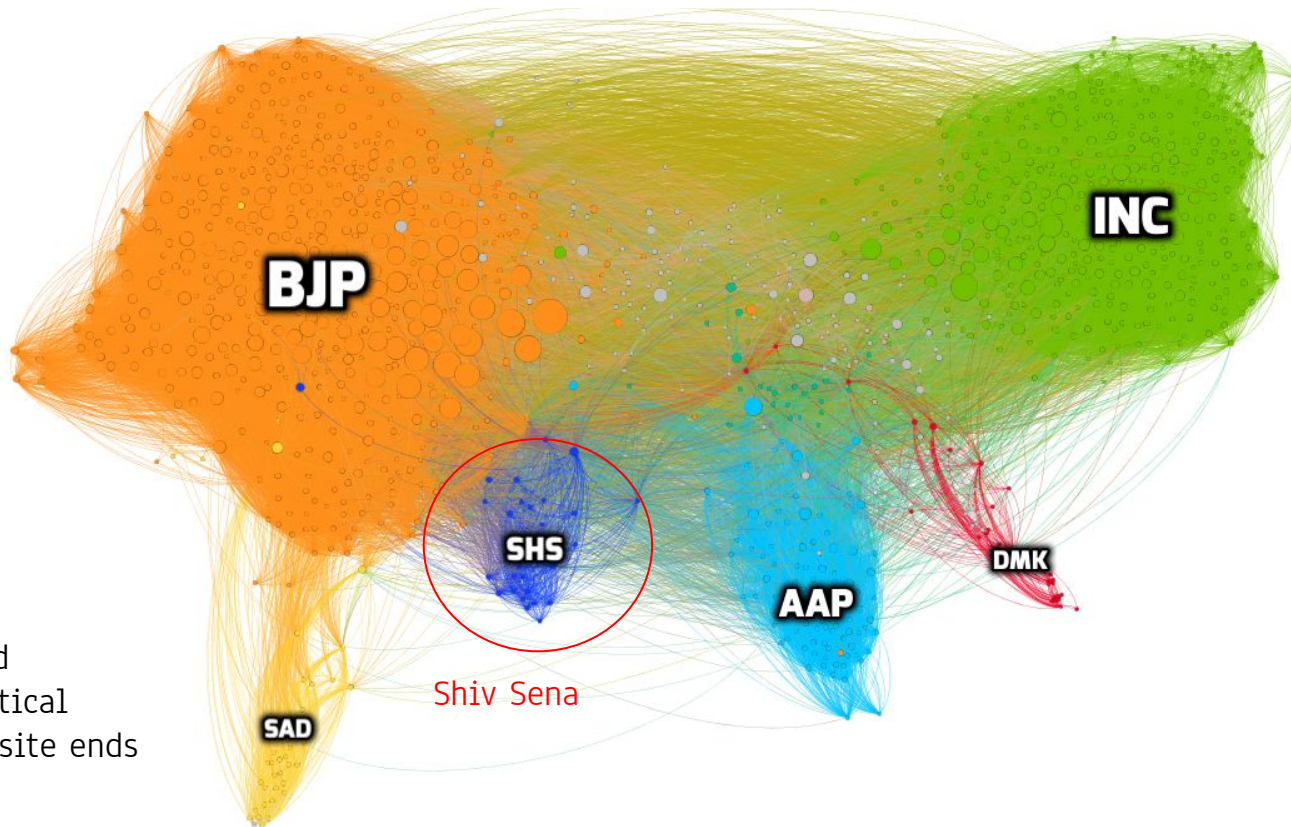


Positioning of
centre-right and
centre-left political
parties on opposite ends
of the network

ANALYSING THE NETWORK VISUALISATION

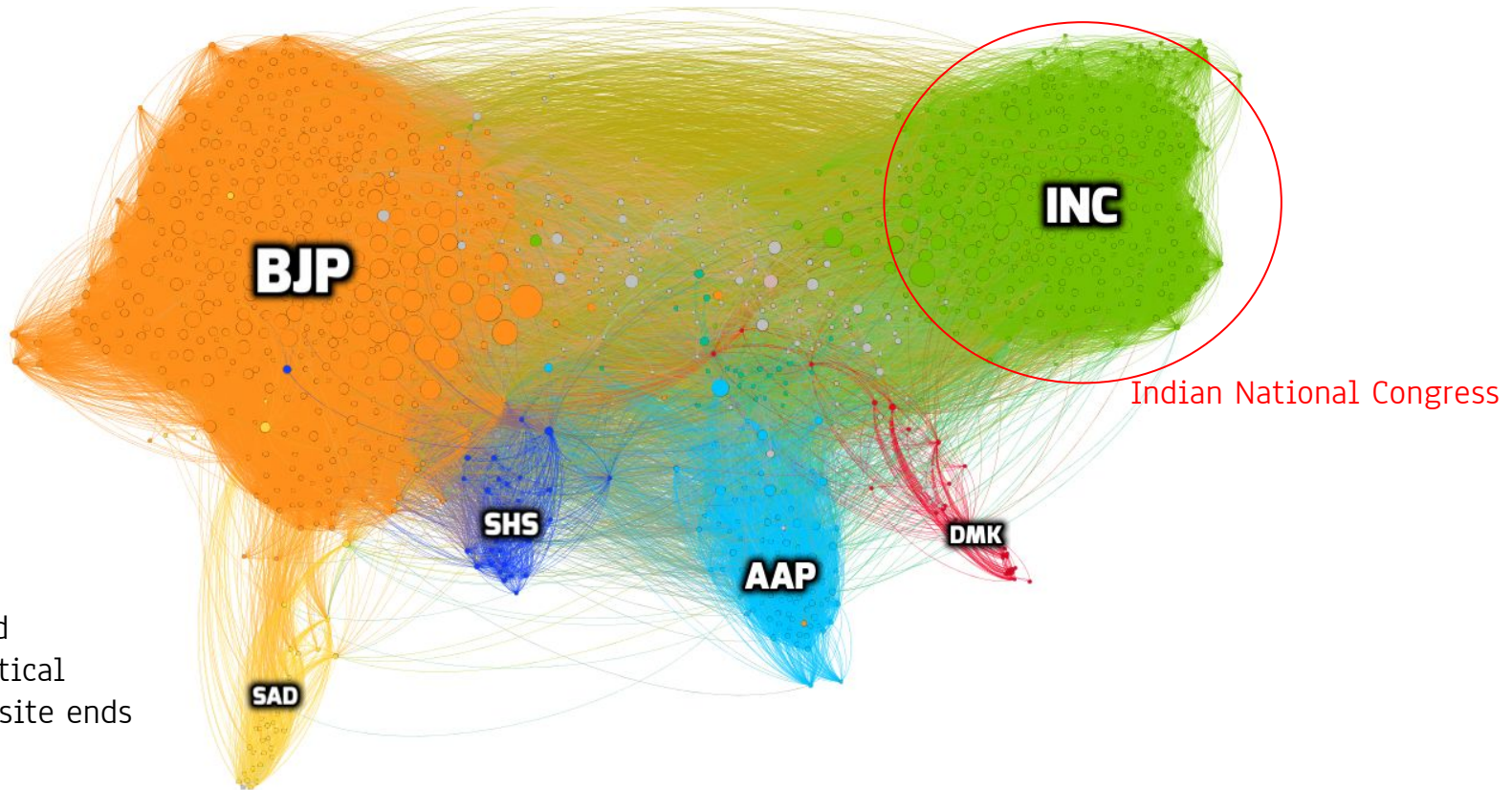


ANALYSING THE NETWORK VISUALISATION

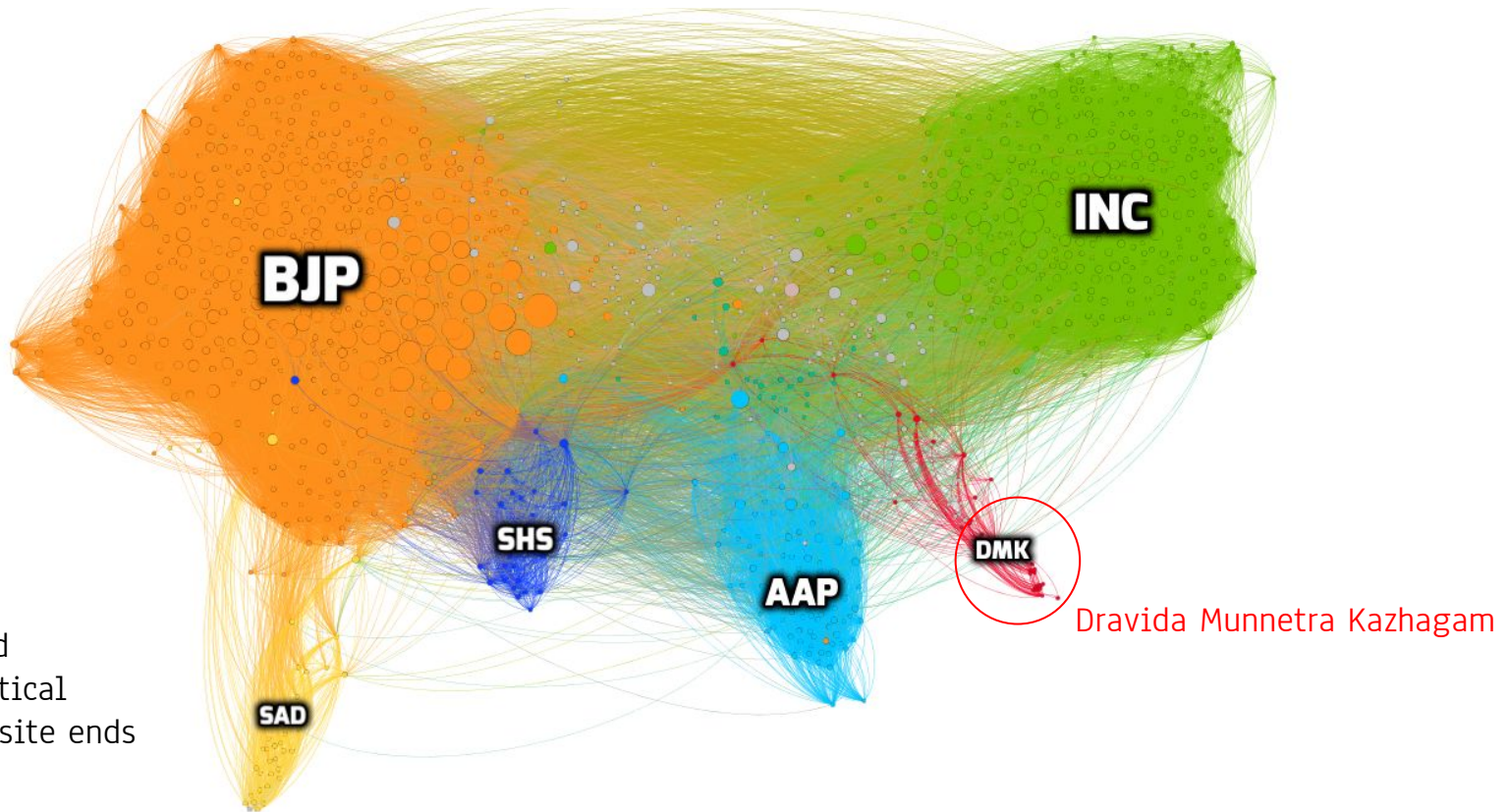


Positioning of
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ANALYSING THE NETWORK VISUALISATION

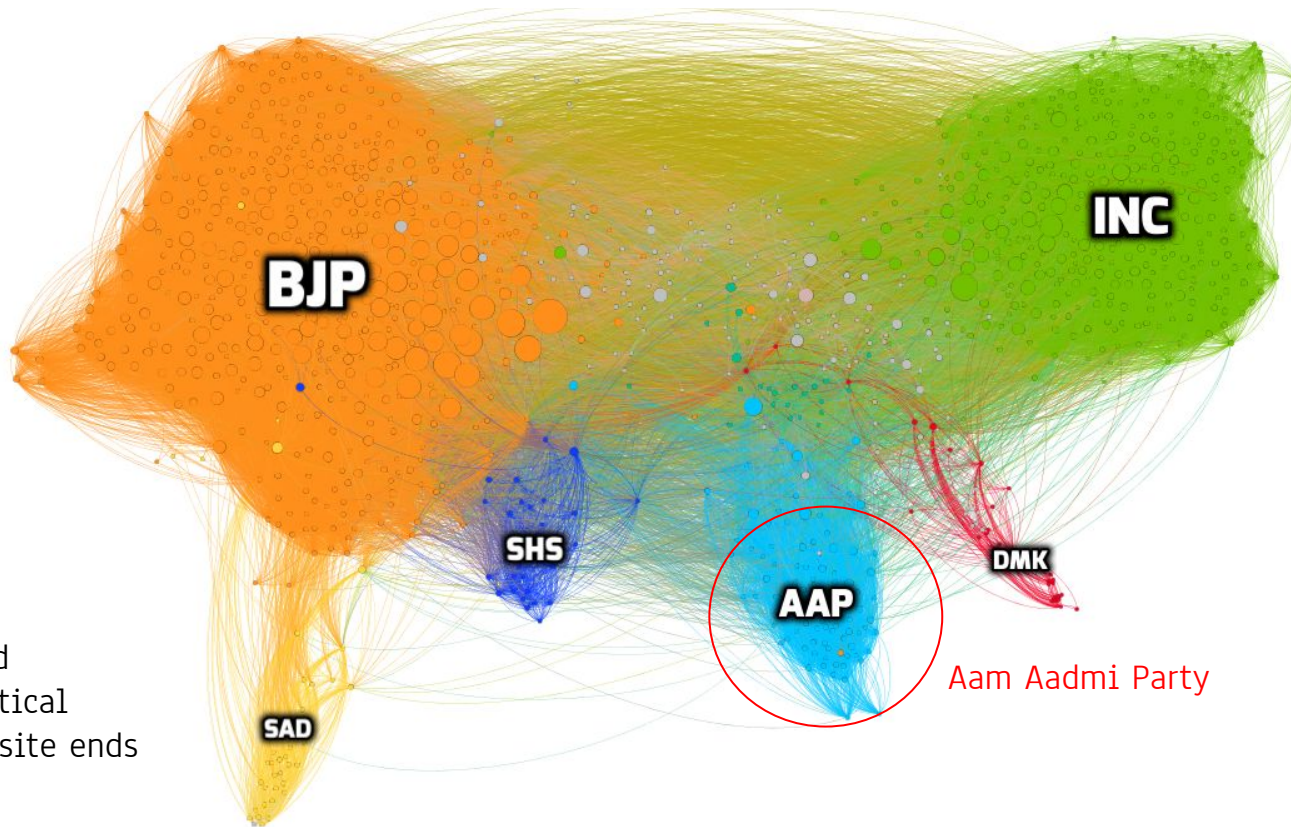


ANALYSING THE NETWORK VISUALISATION



Positioning of
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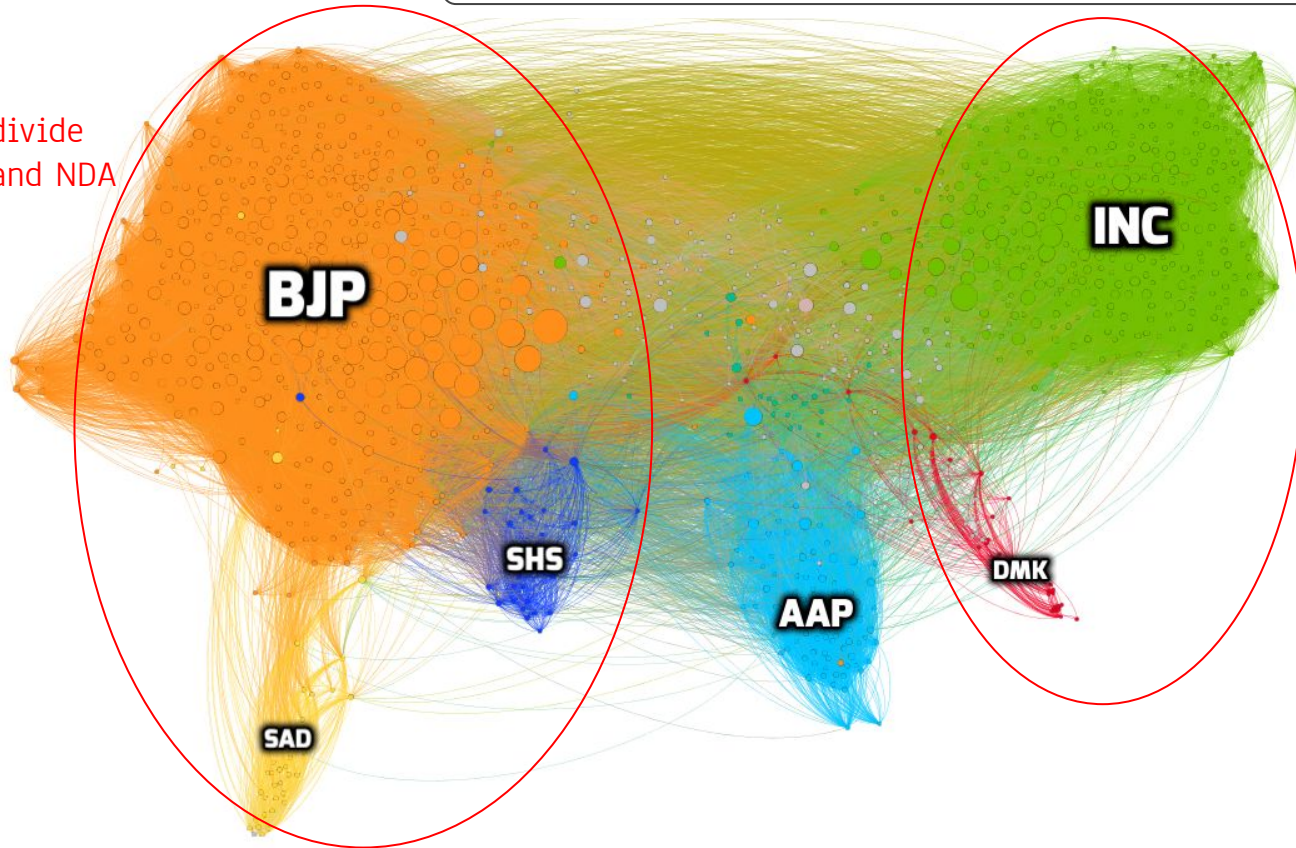
ANALYSING THE NETWORK VISUALISATION



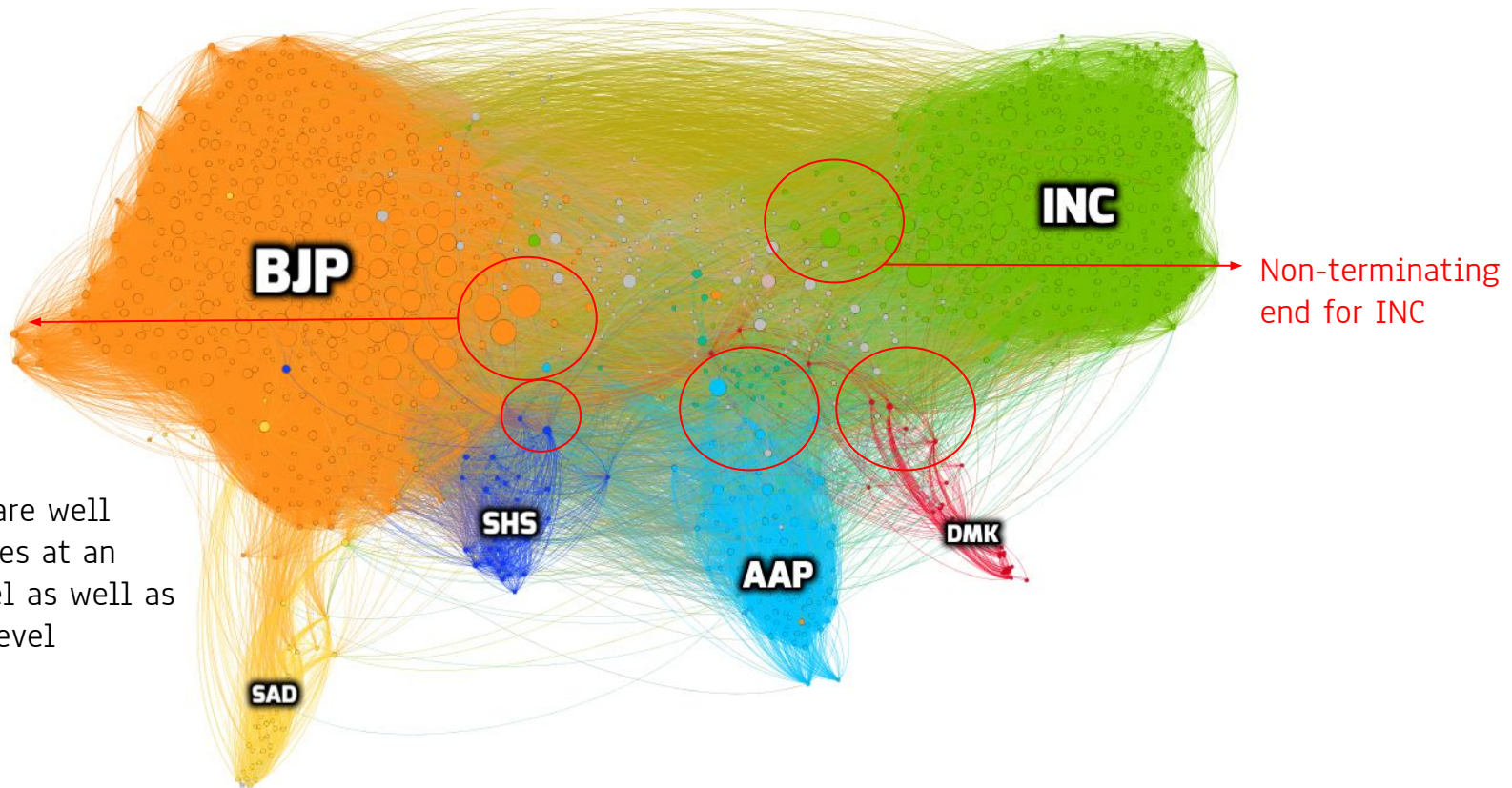
Positioning of
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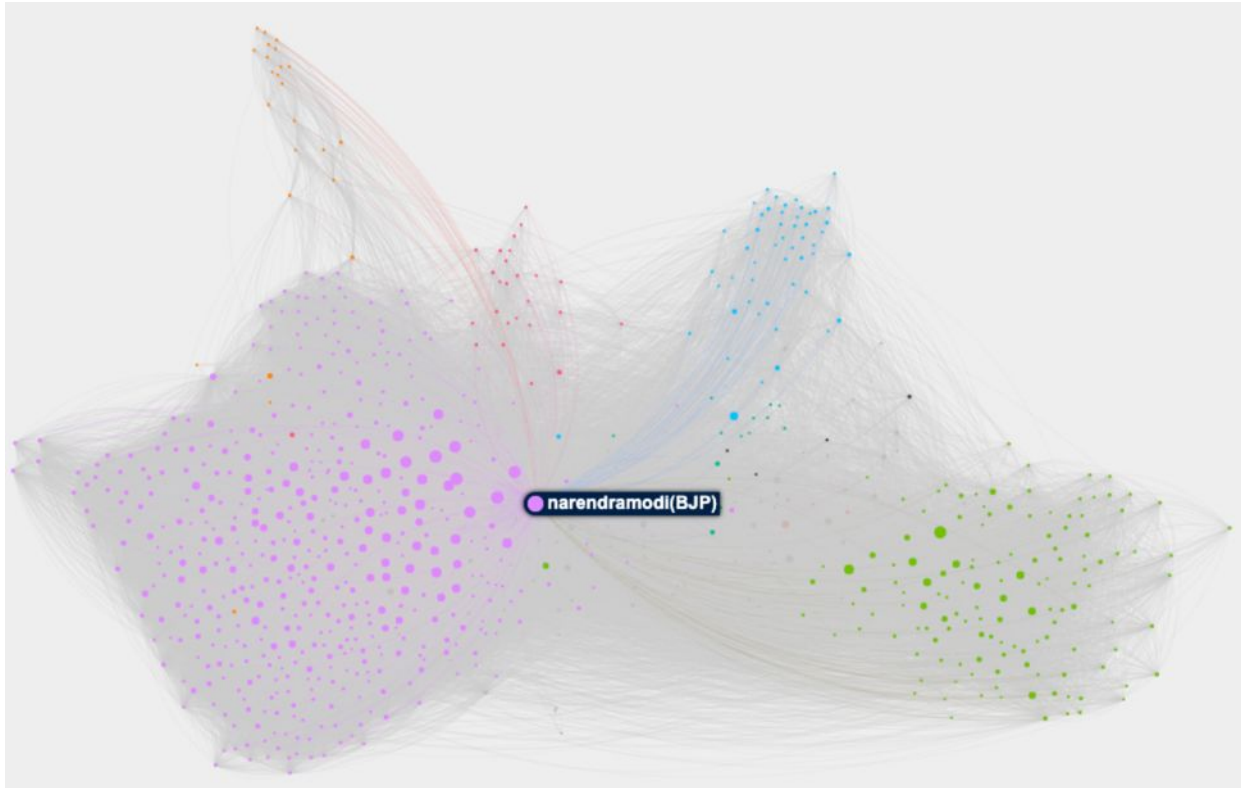
Clear cluster divide
between UPA and NDA



ANALYSING THE NETWORK VISUALISATION

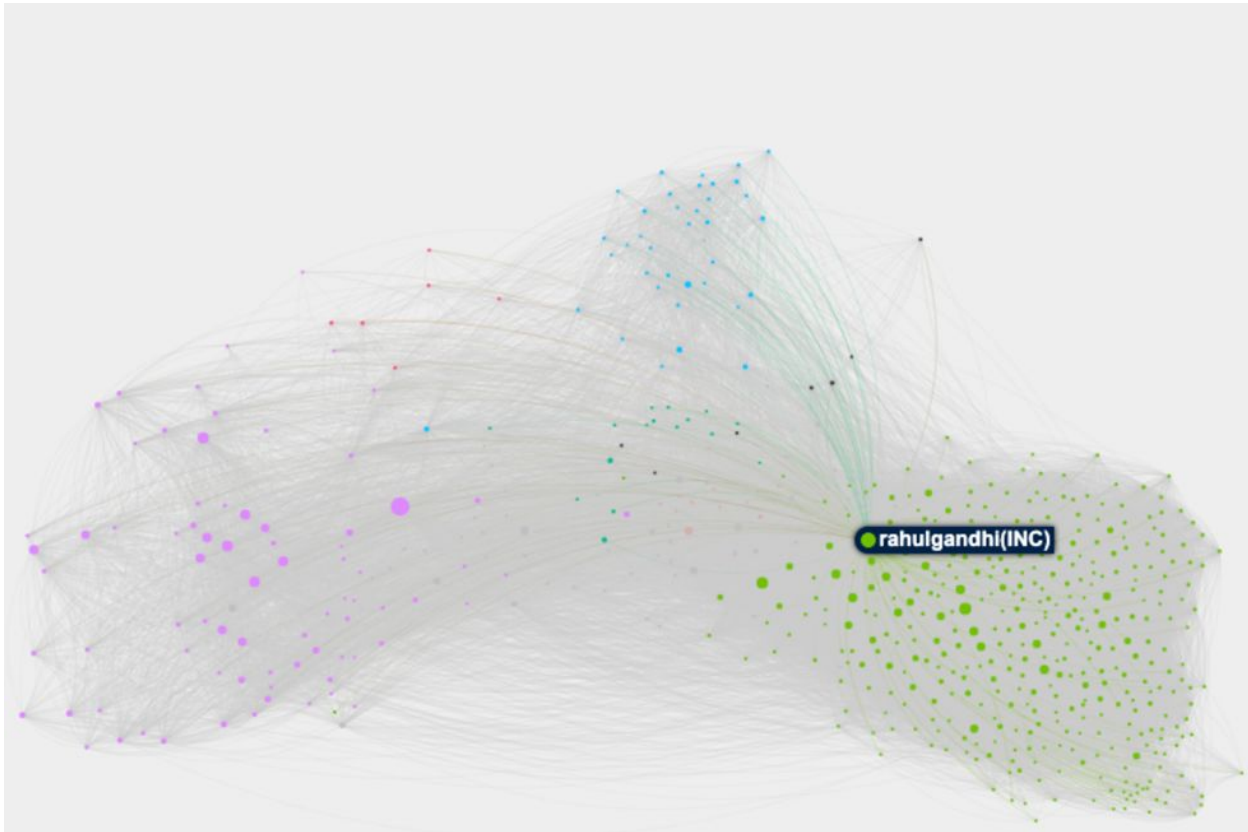


NARENDRA MODI'S NETWORK



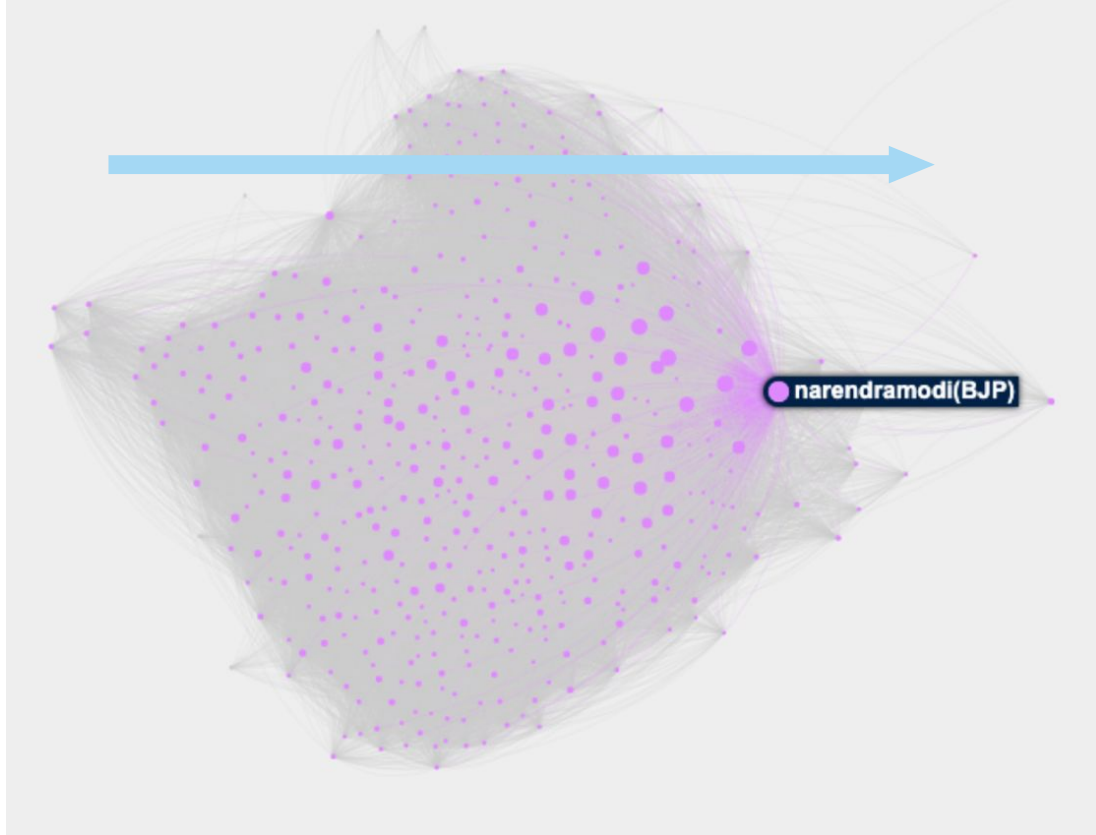
Nodes towards the non-terminating end of the cluster have significant number of individual connections outside their core cluster.

RAHUL GANDHI'S NETWORK



Nodes towards the non-terminating end of the cluster have significant number of individual connections outside their core cluster.

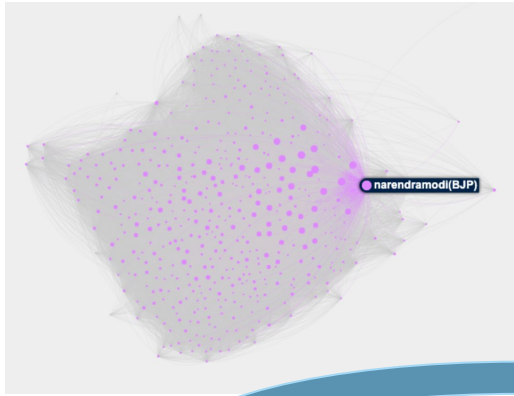
BJP'S CLEAR HIERARCHIES



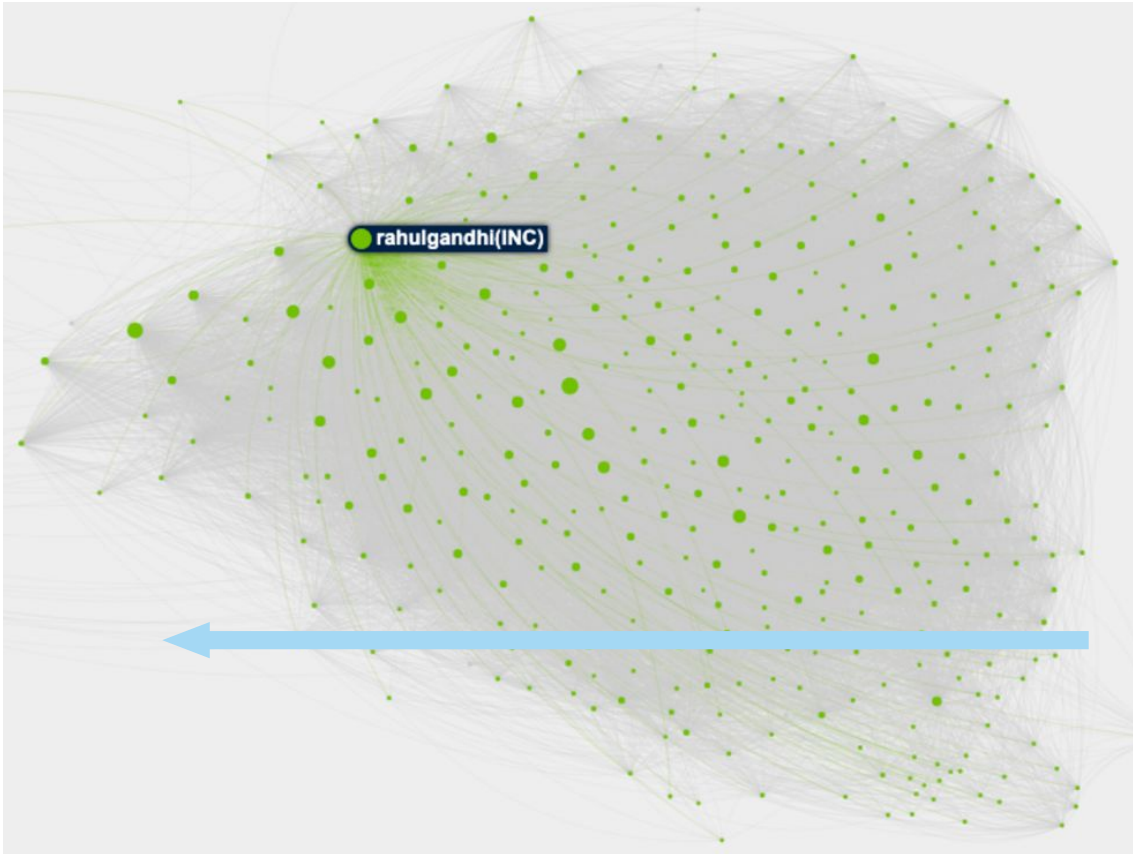
Left to Right

- Node size increases
- Apparent political authority increases
- Social media presence increases
- No node larger than/as large as @narendramodi

BJP'S CLEAR HIERARCHIES



INC'S MISSING HIERARCHIES



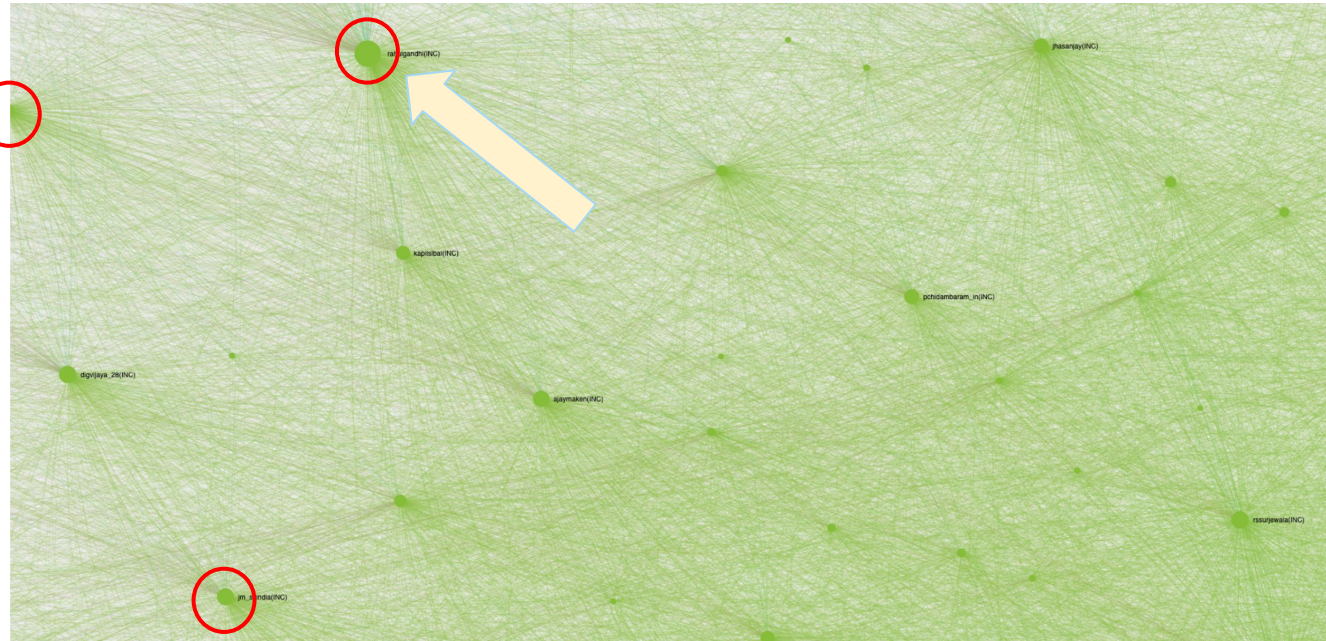
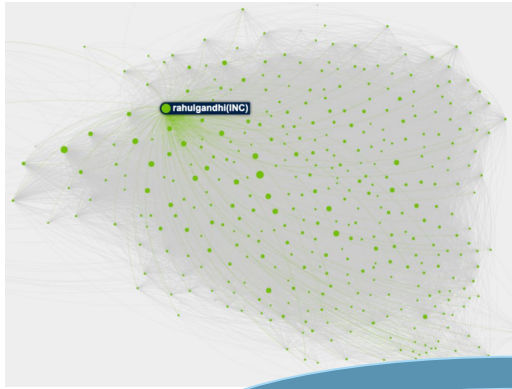
Right to Left

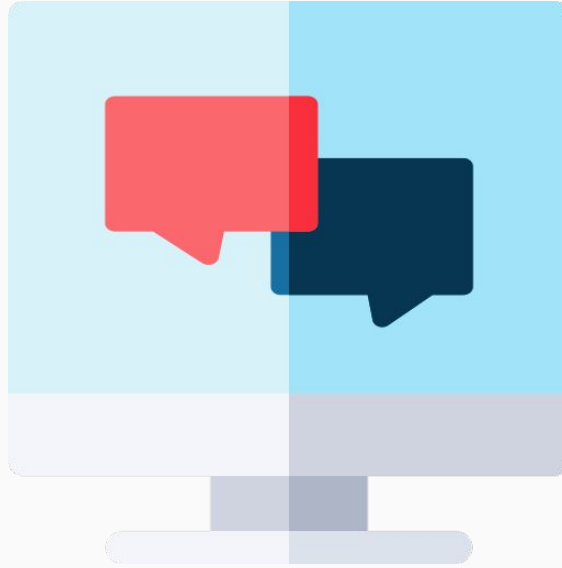
- Node size increases
- Missing Apparent political hierarchy
- There are nodes as large as @rahulgandhi

INC'S MISSING HIERARCHIES

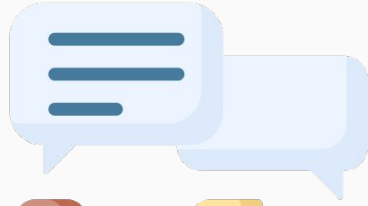
@shashithroor, @jm_scindia (Jyotiraditya Scindia)

- More central to the overall network than @rahulgandhi





DISCOURSE ANALYSIS 2019



DISCOURSE ANALYSIS

DISCOURSE

Discourse refers to how we think and communicate about people, things, the social organization of society, and the relationships among and between all three.

WHY STUDY DISCOURSE

Provide legitimacy for certain kinds of knowledge while undermining others

DISCOURSE ANALYSIS - 2019

- 2019 Lok Sabha Elections
- Analysed on a monthly basis
- Frequence-based tag cloud analysis + TF-IDF
- BJP v/s INC
 - Monthwise, comparative analysis
 - How the narratives were formed
 - How discourses changed over months



BJP

JAN

INC



BJP

FEB

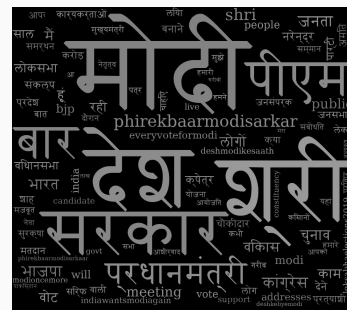
INC



BJP

MAR

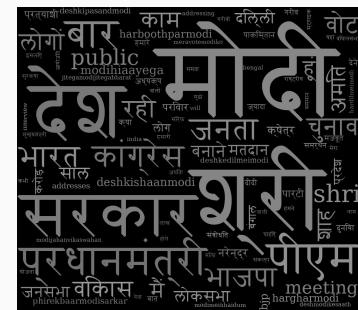
INC



BJP

APR

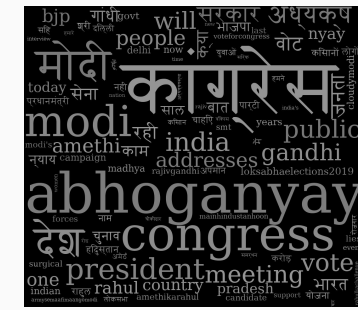
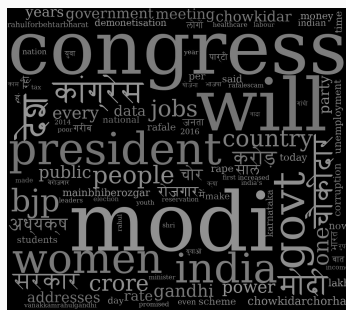
INC



B1P

MAY

TNC



Discourses formed on Twitter by
both the major political parties -
BJP & INC were very divergent.

CONCLUSIONS?

CONCLUSIONS

INC's primary online Campaign

- #AbHogaNyay
 - Curb poverty, create jobs, minimum income guarantee
 - Just one hashtag till the end of the campaign online
 - Socio-political issue

BJP's primary campaign

- #PhirEkBaarModiSarkar
 - Modi's high popularity, coming back to power
 - Multiple hashtags - *#HarBoothParModi, #ModiHiAayega*
 - *Not really an issue*

CONCLUSIONS

INC

- “Chowkidar” campaign
 - March 2019
 - *‘Chowkidar chor hai’*
 - Against PM Modi

BJP

- “Main bhi chowkidar campaign”
 - Began later
 - Gained more popularity
 - Backfired for the congress

CONCLUSIONS

- #LokSabhaElections2019 (Official Twitter Hashtag)
- BJP - 58% Hindi, 30% English, 5% Regional
- INC - 26% Hindi, 61% English, 4% Kannada, 1% All other regional
- BJP handles - more frequent use of "Modi"
- INC handles - more frequent use of "Congress"



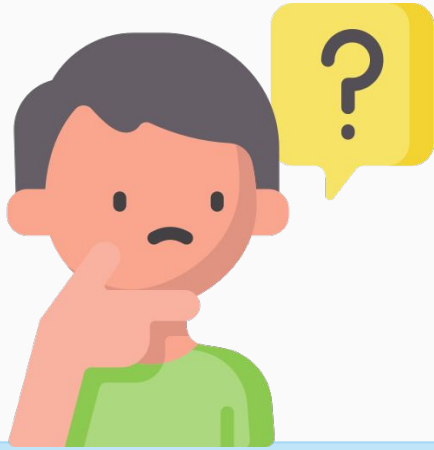
2014 VS 2019

2014 VS 2019

- 2014 & 2019 General Elections
 - Key turning points in the History of India's electoral democracy
 - Major changes for INC and BJP
- Same winning party - BJP
 - Different frameworks
 - Context naturally, and politically different

2014 VS 2019

- 2014
 - INC in power for 10 years
 - BJP main opposition
 - INC - Defend its 10 years of rule
 - AAP - 3rd option
- 2019
 - BJP incumbent
 - INC main opposition
 - No 3rd alternative



QUESTIONS WE TRY TO ANSWER

- How the ruling party and the opposition brought about a change in the discourses and narratives they set in the two elections.
- Did BJP sustain the primary issue of its 2014 campaign in 2019, and did it continue to mobilize on it?

CONCLUSIONS

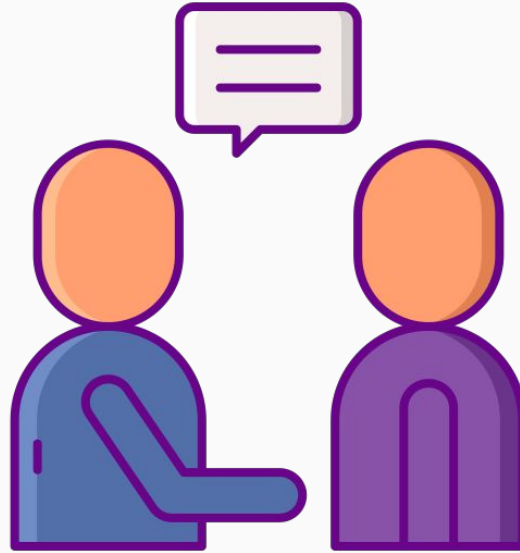
2014 General Elections

- Corruption, women, youth, development - Sustained
- Strong anti-incumbency wave
- Strong discourse for the opposition - BJP
- Jan Lokpal, gas, money, protest etc. - Narratives were formed online

CONCLUSIONS

2019 General Elections

- Absence of politically relevant terms
 - Corruption, development, women, youth etc.
- Brand “Modi”
 - *ModiOnceMore, NamoAgain, IndiaVotesForNam, ModiAaRahaHai, ModiSweep, ModiAaGaya*
- “If not Modi, then who?”
 - *Pappu*; only one major pro-INC hashtag - “#MeraVoteCongressKo”



COMPARISON WITH ON-GROUND SURVEYS

CSDS & LOKNITI

- Center for the Study of Developing Societies
- Lokniti
- Mood of the nations
 - Round 1: 1 - 15 May 2017
 - Round 2: 7 - 20 January 2018
 - Round 3: 28 April - 17 May 2019

CONCLUSIONS

- Mood of the Nation - Government did not fulfil the promises it had made during 2014
 - Unemployment, farmer distress in 2017
 - Not raised in 2019; Major area of criticism
 - Traders and farmers unhappy with GST and demonetisation

CONCLUSIONS

- Stark contrast
- 3 Mood of the Nations - Decrease in Modi's popularity; Twitter proves otherwise
 - Brand "Modi" in 2019
 - modioncemore, mainbhichowkidar, indiavotesfornamo, wewantchowkidaar, modiaarahahai
 - Network Analysis
- 3 Mood of the Nations - Strong anti-incumbency wave
 - Online discourse & results - Otherwise

PRELIMINARY ANALYSIS

- > BJP - Far greater reach
- > Number of handles, activity or influence

NETWORK ANALYSIS

- > Clusters synonymous to real world; Alliance
- > BJP - Apparent political hierarchy



DISCOURSE ANALYSIS

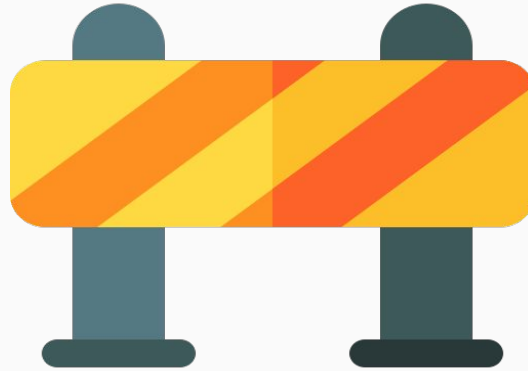
- > INC BJP - Very Divergent
- > Socio-political issues V/S Modi brand

2019 V/S 2014 ONLINE

- > Politically relevant issues absent
- > Modi brand V/S Anti-Modi brand

TWEETS V/S SURVEYS

- > Surveys - Anti incumbent wave
- > Tweets - Pro incumbent wave; Election results



LIMITATIONS AND FUTURE WORK

LIMITATIONS

I

Code
Mixed Data

II

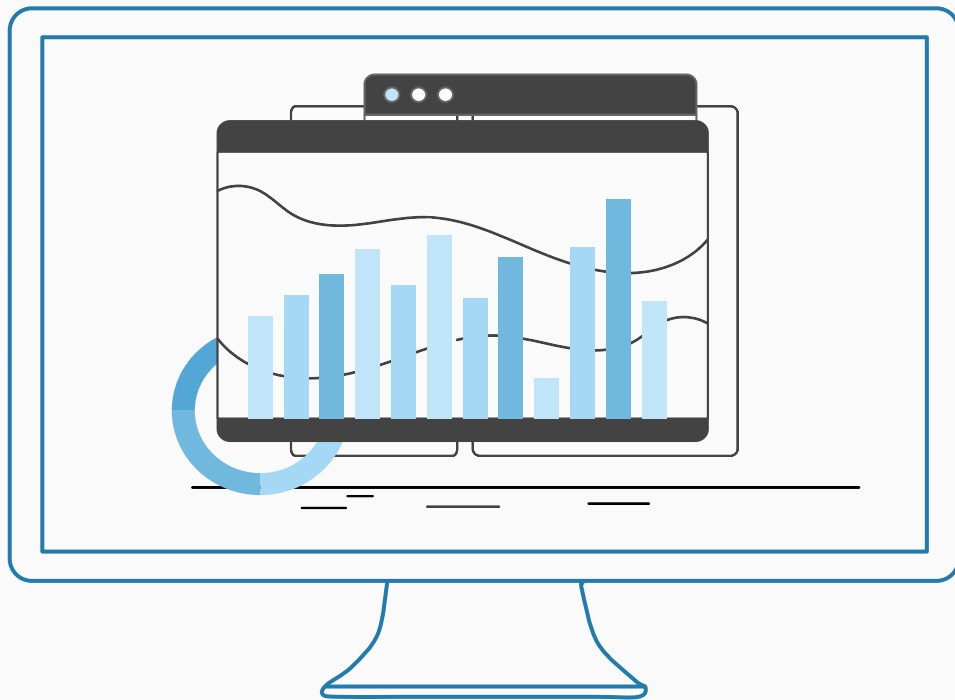
Code
switching

III

LDA,
TF-IDF,
NER, ABSA

FUTURE WORK

- Discourse Analysis + Network Analysis
 - Echo Chambers
 - How information flows during an election campaign
 - How are the narratives change and what factors (or actors) lead to this change
- Dataset consists of Tweets of Media handles
 - What role does the media play in shaping narratives
 - Neutral / Biased



bit.ly/elections19



Thank you

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